STAT210B Lecture 2 Sub-Gaussian Random Variables

Logic

在上节课中, 我们知道了 Markov 不等式可以作用于 $e^{\lambda X}$ (Chernoff's method), 由此可以得到指数尾界

若 X 的 tail 比 Gaussian 更 light, 则我们可以利用 Gaussian 的指数尾界来 bound X, 对于这一类拥有 comparable tail behavior 的 X, 我们称其为 sub-Gaussian random variables

1 Sub-Gaussian Random Variables

1.1 Gaussian random variable 的 MGF

令 $X \sim \mathcal{N}(0, \sigma^2)$, 则 $M_X(\lambda) = \mathbb{E}[\exp(\lambda X)] = \exp\left(\frac{\lambda^2\sigma^2}{2}\right)$

Proof

$$
\begin{aligned}
\mathbb{E}[\exp(\lambda X)] &= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma}\exp(\lambda x)\cdot\exp\left(-\frac{x^2}{2\sigma^2}\right)dx \\
&= \exp\left(\frac{\lambda^2\sigma^2}{2}\right)\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{x^2}{2\sigma^2}+\lambda x-\frac{\lambda^2\sigma^2}{2}\right)dx \\
&= \exp\left(\frac{\lambda^2\sigma^2}{2}\right)\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{1}{2\sigma^2}(x-\lambda\sigma^2)^2\right)dx \\
&= \exp\left(\frac{\lambda^2\sigma^2}{2}\right)\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{y^2}{2\sigma^2}\right)dy \quad (\text{令 } y = x-\lambda\sigma^2) \\
&= \exp\left(\frac{\lambda^2\sigma^2}{2}\right)
\end{aligned}
$$

Example: Chernoff's Method for Gaussian Random Variable

对于 $X \sim \mathcal{N}(0, \sigma^2)$, 使用 Chernoff's method, 则有

$$
\mathbb{P}(Z \geq t) \leq \inf_{\lambda>0}\frac{M_Z(\lambda)}{\exp(\lambda t)} = \inf_{\lambda>0}\exp\left(\frac{\lambda^2\sigma^2}{2}-\lambda t\right) = \exp\left(-\frac{t^2}{2\sigma^2}\right) \quad (\text{令 } \lambda = \frac{t}{\sigma^2})
$$

1.2 Definition: Sub-Gaussian random variable

令 X 为 **zero-mean** random variable,

则 X 为 sub-Gaussian with (variance) parameter $\sigma^2 > 0$, 若

$$
M_X(\lambda) = \mathbb{E}[\exp(\lambda X)] \leq \exp\left(\frac{\lambda^2\sigma^2}{2}\right), \quad \forall\lambda\in\mathbb{R}
$$

Remark: $-X$ 也为 sub-Gaussian

若 X 为 sub-Gaussian with parameter σ, 则 $-X$ 也为 sub-Gaussian with parameter σ^2, 因为

$$
\mathbb{E}e^{\lambda(-X)} = \mathbb{E}e^{(-\lambda)X} \leq \exp\left(\frac{(-\lambda)^2\sigma^2}{2}\right) = \exp\left(\frac{\lambda^2\sigma^2}{2}\right), \quad \forall\lambda\in\mathbb{R}
$$

Example: Gaussian Random Variable

若 $Z \sim \mathcal{N}(0, \sigma^2)$, 则 Z 为 sub-Gaussian with variance parameter σ^2, 由于 $\mathbb{E}[e^{\lambda Z}] = e^{\lambda^2\sigma^2/2}$

Example: Non-Gaussian Sub-Gaussian Random Variable

考虑 Rademacher random variable, 即 $\varepsilon = \pm 1$ with equal probability, 则有

$$\mathbb{E}[\exp(\lambda\varepsilon)] = \frac{1}{2}\exp(\lambda) + \frac{1}{2}\exp(-\lambda) \leq \exp\left(\frac{\lambda^2}{2}\right)$$

因此 Rademacher random variable 为 sub-Gaussian with parameter 1

Remark: 关于 Zero-mean 的解释

若希望 X 的 MGF 被 $\exp(\lambda^2\sigma^2/2)$ bound, 则要求

$$\begin{aligned}\mathbb{E}[\exp(\lambda X)] &= 1 + \mathbb{E}[\lambda X] + \mathbb{E}\left[\frac{\lambda^2 X^2}{2}\right] + \cdots \\ &\leq 1 + \frac{\lambda^2\sigma^2}{2} + \cdots\end{aligned}$$

即要求 $\mathbb{E}[X] = 0$

对于 X with $\mathbb{E}[X] \neq 0$, 我们有

$$\begin{aligned}\mathbb{E}[\exp(\lambda X)] &= \mathbb{E}[\exp(\lambda(X - \mathbb{E}[X]) + \lambda\mathbb{E}[X])] \\ &= \exp(\lambda\mathbb{E}[X]) \cdot \mathbb{E}[\exp(\lambda(X - \mathbb{E}[X]))]\end{aligned}$$

其可以视作 centered X 的 MGF 的常数倍, 因此即便不 center, 也不会显著影响 tail behavior

1.3 Property: Sub-Gaussian Random Variable 的性质

若 X 为 sub-Gaussian random variable,

则对 $\forall t \geq 0$,

$$\mathbb{P}(X \geq t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

and

$$\mathbb{P}(|X| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sigma^2}\right)$$

Proof: Property 1

使用 Chernoff's method 即可:

$$\mathbb{P}(X \geq t) \leq \inf_{\lambda > 0} \frac{M_X(\lambda)}{\exp(\lambda t)} \leq \inf_{\lambda > 0} \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda t\right) = \exp\left(-\frac{t^2}{2\sigma^2}\right) \quad (令\ \lambda = \frac{t}{\sigma^2})$$

Proof: 使用 Property 1 证明 Property 2

使用 union bound 即可:

$$\begin{aligned}\mathbb{P}(|X| \geq t) &= \mathbb{P}((X \geq t) \cup (X \leq -t)) \\ &= \mathbb{P}((X \geq t) \cup (-X \geq t)) \\ &\leq \mathbb{P}(X \geq t) + \mathbb{P}(-X \geq t) \\ &= \exp\left(-\frac{t^2}{2\sigma^2}\right) + \exp\left(-\frac{t^2}{2\sigma^2}\right) \\ &= 2\exp\left(-\frac{t^2}{2\sigma^2}\right)\end{aligned}$$

1.4 Proposition: Sub-Gaussian 的等价定义

令 random variable X 满足 $\mathbb{E}[X]=0$,

则以下 conditions 等价 (with parameters k_1, k_2, k_3, k_4, k_5 being the same up to universal multiplicative constant factors):

Condition 1 (Definition):

存在 $k_1>0$, 使得对 $\forall\lambda\in\mathbb{R}$,

$$\mathbb{E}[\exp(\lambda X)] \le \exp(\lambda^2 k_1^2)$$

Condition 2 (Property):

存在 $k_2>0$, 使得对 $\forall t\ge 0$,

$$\mathbb{P}(|X|\ge t) \le 2\exp\left(-\frac{t^2}{k_2^2}\right)$$

Condition 3:

存在 $k_3>0$, 使得对 $\forall p\ge 1$,

$$\|X\|_{L_p} = (\mathbb{E}[|X|^p])^{\frac{1}{p}} \le k_3\sqrt{p}$$

Condition 4:

存在 $k_4>0$, 使得对 $\forall\lambda \text{ with } |\lambda|\le 1/k_4$,

$$\mathbb{E}[\exp(\lambda^2 X^2)] \le \exp(\lambda^2 k_4^2)$$

Condition 5:

存在 k_5,

$$\mathbb{E}\left[\exp\left(\frac{X^2}{k_5^2}\right)\right] \le 2$$

Remark:

k_1, k_2 being the same up to universal multiplicative constant factors 表示: 存在 $c_1, c_2>0$ (与 X, k_i 独立), 使得 $c_1 k_1 < k_2 < c_2 k_1$

Proof: Condition 1 $\rightarrow$ Condition 2

类似于证明 Property, 使用 Chernoff's method 即可

Proof: Condition 2 $\rightarrow$ Condition 3

对于 non-negative random variable $Y\ge 0$, 有以下和 tail probability 相关的性质:

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y\ge t)\,dt$$

For simplicity, 令 $k_2=1$, 因此有

$$
\begin{aligned}
\mathbb{E}[|X|^p] &= \int_0^\infty \mathbb{P}(|X|^p \geq t)\,dt \\
&= \int_0^\infty \mathbb{P}(|X| \geq u) p u^{p-1}\,du \quad (\text{令 } t = u^p, dt = pu^{p-1}du) \\
&\leq \int_0^\infty 2\exp(-u^2) p u^{p-1}\,du \quad (\text{使用 Property 2}) \\
&= \int_0^\infty \frac{2\exp(-w) p w^{\frac{p}{2}-1}}{2}\,dw \quad (\text{令 } u^2 = w, dw = 2udu) \\
&= p\,\Gamma\left(\frac{p}{2}\right)
\end{aligned}
$$

对于 Gamma function, 使用 standard bounds $\Gamma(x) \leq 3x^x, \forall x \geq \frac{1}{2}$, 有

$$
\begin{aligned}
&\mathbb{E}[|X|^p] \leq p\left(\frac{p}{2}\right)^{\frac{p}{2}}, \quad p \geq 1 \\
\Rightarrow \quad &(\mathbb{E}[|X|^p])^{\frac{1}{p}} \leq p^{\frac{1}{p}}(\sqrt{p}), \quad p \geq 1
\end{aligned}
$$

1.5 Proposition: Hoeffding Lemma (Bounded Random Variables 为 Sub-Gaussian)

令:

- X 为 random variable
- $\mathbb{E}[X] = 0$
- $X \in [a, b]$ almost surely

则对 $\forall \lambda \in \mathbb{R}$,

$$
\mathbb{E}[\exp(\lambda X)] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right)
$$

即 X 为 sub-Gaussian random variable with variance parameter $\frac{(b-a)^2}{4}$

Proof: 一个 simplified proof

我们首先证明一个 simplified version:

$$
\mathbb{E}[\exp(\lambda X)] \leq \exp\left(\frac{\lambda^2(b-a)^2}{2}\right)
$$

使用 idea of symmetrization, 令 X' 为 independent copy of X, 则

$$
\begin{aligned}
\mathbb{E}[\exp(\lambda X)] &= \mathbb{E}[\exp(\lambda(X - \mathbb{E}[X']))] \\
&\leq \mathbb{E}[\exp(\lambda(X - X'))] \\
&= \frac{1}{2}\mathbb{E}[\exp(\lambda(X - X'))] + \frac{1}{2}\mathbb{E}[\exp(\lambda(X' - X))] \\
&\leq \mathbb{E}[\exp(\frac{\lambda^2}{2}(X - X')^2)] \quad (\text{比较 Taylor expansion, 有 } \frac{1}{2}\exp(x) + \frac{1}{2}\exp(-x) \leq \exp\left(\frac{x^2}{2}\right)) \\
&\leq \exp(\frac{\lambda^2}{2}(b-a)^2)
\end{aligned}
$$

Remark: 关于 constant

通常情况下, 我们仅需要 "bounded random variables 为 sub-Gaussian" 的结论, 因此 constant 是否最优往往并不重要

Logic

MGF 的一个重要优点是它与独立性相容, 这可以帮助我们求出 sums of independent random variables 的 concentration bounds

2 Sums of Independent Sub-Gaussians

2.1 Proposition: Sum of Sub-Gaussians 的性质

令:

- $X_1, \dots, X_n$ 为 independent random variables (不要求同分布)
- $\mathbb{E}[X_i] = \mu_i, \quad \forall i \in \{1, \dots, n\}$
- $X_i - \mu_i$ 为 sub-Gaussian with variance parameter $\sigma_i^2, \quad \forall i \in \{1, \dots, n\}$

则对 $\forall t \geq 0$,

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mu_i) \geq t\right) \leq \exp\left(-\frac{t^2}{2\sum_{i=1}^{n}\sigma_i^2}\right)$$

and

$$\mathbb{P}\left(\left|\sum_{i=1}^{n}(X_i - \mu_i)\right| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2\sum_{i=1}^{n}\sigma_i^2}\right)$$

Proof

由于 independence, sum of sub-Gaussians 的 MGF 有以下 bound:

$$\begin{aligned}\mathbb{E}\left[\exp\left(\lambda\left(\sum_{i=1}^{n}(X_i - \mu_i)\right)\right)\right] &= \prod_{i=1}^{n}\mathbb{E}[\exp(\lambda(X_i - \mu_i))] \\ &\leq \prod_{i=1}^{n}\exp(\lambda^2\sigma_i^2/2) \\ &= \exp\left(\lambda^2\left(\sum_{i=1}^{n}\sigma_i^2\right)/2\right)\end{aligned}$$

由 Chernoff's method, 有

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mu_i) \geq t\right) \leq \inf_{\lambda}\exp\left(\lambda^2\sum_{i=1}^{n}\sigma_i^2/2 - \lambda t\right) = \exp\left(-\frac{t^2}{2\sum_{i=1}^{n}\sigma_i^2}\right)$$

使用 symmetry (对 $-\sum_{i=1}^{n}(X_i - \mu_i)$ 使用相同的 argument) 和 union bound, 有

$$\mathbb{P}\left(\left|\sum_{i=1}^{n}(X_i - \mu_i)\right| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2\sum_{i=1}^{n}\sigma_i^2}\right)$$

Remark: Sum of Sub-Gaussians 仍为 Sub-Gaussian

在上述证明中, 我们证明了

$$\mathbb{E}\left[\exp\left(\lambda\left(\sum_{i=1}^{n}(X_i - \mu_i)\right)\right)\right] \leq \exp\left(\lambda^2\left(\sum_{i=1}^{n}\sigma_i^2\right)/2\right)$$

因此 sum of sub-Gaussians 为 sub-Gaussian with variance parameter $\sum_{i=1}^{n}\sigma_i^2$

2.2 Proposition: Hoeffding's inequality (Sum of Bounded Random Variables 为 Sub-Gaussian)

令:

- $X_1, \ldots, X_n$ 为 independent random variables (不要求同分布)
- $\mathbb{E}[X_i] = \mu_i, \quad \forall i \in \{1, \ldots, n\}$
- $X_i \in [a_i, b_i]$ almost surely, $\quad \forall i \in \{1, \ldots, n\}$

则对 $\forall t \geq 0$,

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mu_i) \geq t\right) \leq \exp\left(-\frac{t^2}{2\sum_{i=1}^{n}(b_i - a_i)^2}\right)$$

and

$$\mathbb{P}\left(\left|\sum_{i=1}^{n}(X_i - \mu_i)\right| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2\sum_{i=1}^{n}(b_i - a_i)^2}\right)$$

Proof

根据 Hoeffding's lemma (simplified version), 有

$$\mathbb{E}[\exp(\lambda(X_i - \mu_i))] \leq \exp\left(\frac{\lambda^2(b_i - a_i)^2}{2}\right)$$

即 $X_i - \mu_i$ 为 sub-Gaussian with variance parameter $\sigma_i^2 = (b_i - a_i)^2$,

随后使用 sum of sub-Gaussians 的性质即可得到结论

Example: Average of Random Signs

令 $\varepsilon_1, \ldots, \varepsilon_n$ 为 independent Rademacher random variable, 则有 $\varepsilon_i \in [-1, 1]$ and $\mathbb{E}[\varepsilon_i] = 0$,
因此使用 Hoeffding's inequality, 对 $\forall t \geq 0$,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\right| \geq t\right) \leq 2\exp\left(-\frac{nt^2}{8}\right)$$

即对于 $\forall \delta \in (0, 1)$, 有 $1 - \delta$ 的 probability,

$$\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\right| \leq \sqrt{\frac{8\log(2/\delta)}{n}}$$

Remark: Comparing with Chebyshev's inequality

对于 Rademacher random variable, 有

$$\operatorname{Var}\left[\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\right] = \frac{1}{n^2}\operatorname{Var}\left[\sum_{i=1}^{n}\varepsilon_i\right] = \frac{1}{n}$$

使用 Chebyshev's inequality, 有

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\right| \geq t\right) \leq \frac{1}{nt^2}$$

即对于 $\forall \delta \in (0, 1)$, 有 $1 - \delta$ 的 probability,

$$\left|\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i\right| \leq \frac{1}{\sqrt{n\delta}}$$

若 $\delta = 0.001$, 则此时考虑比较

$$\sqrt{8\log 2000} \approx 7.798 \quad \left(\sqrt{8\log(2/\delta)}\right) \quad \text{v.s.} \quad \sqrt{1000} \approx 31.623 \quad \left(\frac{1}{\sqrt{\delta}}\right)$$